

IT Asset Request Automation using ServiceNow

This project automates IT asset request and provisioning workflows using ServiceNow Flow Designer and Service Catalog.

1. Project Overview

The IT Asset Request Automation project is developed using the ServiceNow platform to automate IT asset request and provisioning workflows. The system allows employees to request IT assets such as laptops and automatically routes approvals, generates catalog tasks, and tracks fulfillment stages.

2. Objectives

- Automate IT asset requests
- Implement approval workflows
- Retrieve catalog variables dynamically
- Generate catalog tasks automatically
- Improve IT fulfillment efficiency
- Provide visibility using request stages

3. Technologies Used

- ServiceNow
- Flow Designer
- Service Catalog
- Catalog Tasks

4. Modules Used

- Service Catalog
- Flow Designer
- Requested Items (RITM)
- Catalog Tasks (SCTASK)

5. Workflow

- Create a Catalog Item (Request New Laptop)
- Attach it to Flow Designer
- Retrieve Catalog Variables
- Configure Dynamic Approvals based on variables
- Set Stages for visibility

Simply,

Employee → Service Catalog → Approval → Catalog Task → IT Team → Fulfillment

6. Catalog Item Creation

Created a catalog item named '**Request New Laptop**' with variables such as Laptop Type, Cost Center, and Justification.

Laptop Type – Multiple Choices like Standard, Gaming, High Performance

Cost Center - Multiple Choices in Currency

Justification – Single Line Text

The screenshot shows the 'Catalog Item - Request New Laptop' page in ServiceNow. The page includes a header with navigation links and a toolbar with 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue information box states: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.' The form fields include: Name (Request New Laptop), Application (Global), Active (checked), Fulfillment automation level (Unspecified), Category (Services), State (None), Checked out (None), and Owner (System Administrator). Below the form, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Item Details' tab is active, showing a 'Short description' field and a 'Description' field with a rich text editor.

The screenshot shows the 'Variable - Laptop Type' page in ServiceNow. The page includes a header with navigation links and a toolbar with 'Copy', 'Update', and 'Delete'. A blue information box states: 'Specify the Question that explains the options available to the end user when ordering the item'. The form fields include: * Question (Laptop Type), * Name (laptop_type), Conversational label, and Tooltip. Below the form, there are buttons for 'Copy', 'Update', and 'Delete'. The 'Related Links' section includes 'Run Point Scan' and 'SN UPIB Versions (2)'. The 'Question Choices' section shows a table with columns: Question, Text, Value, Order, and Inactive. The table contains five rows of data for the 'Laptop Type' question.

Question	Text	Value	Order	Inactive
Laptop Type	Standard	Standard	100	false
	Gaming	Gaming	200	false
	2-in-1	2-in-1	300	false
	Ultrabooks	Ultrabooks	400	false
	High Performance	High Performance	500	false

servicenow

Variable - Laptop Cost

Question Annotation Type Specifications Default Value Auto-populate Permission Availability

Specify the Question that explains the options available to the end user when ordering the item

* Question: Laptop Cost

* Name: laptop_cost

Conversational label

Tooltip

Copy Update Delete

Related Links

Run Print Scan

SN URGENT Versions (2)

Question Choices

Text	Value	Order	Inactive
\$500	\$50	100	false
\$700	\$700	200	false
\$1000	\$1000	300	false
\$1200		400	false

Insert a new row...

1 to 4 of 4

7. Flow Designer Implementation

The flow triggers automatically when a request is submitted. It retrieves catalog variables, generates approvals, updates request stages, and creates catalog tasks.

Step 1: Trigger flow when Requested Item is created.

Step 2: Retrieve catalog variables.

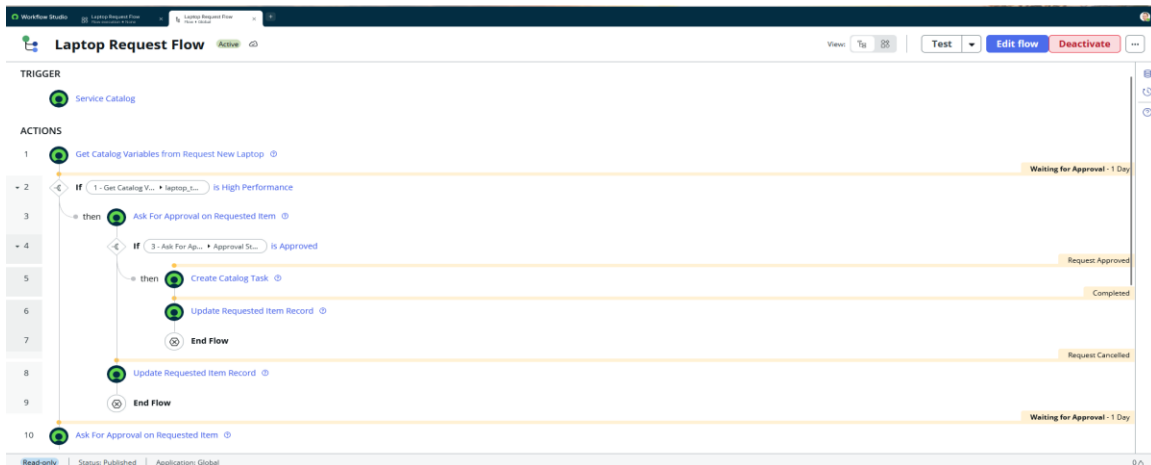
Step 3: Configure approval conditions.

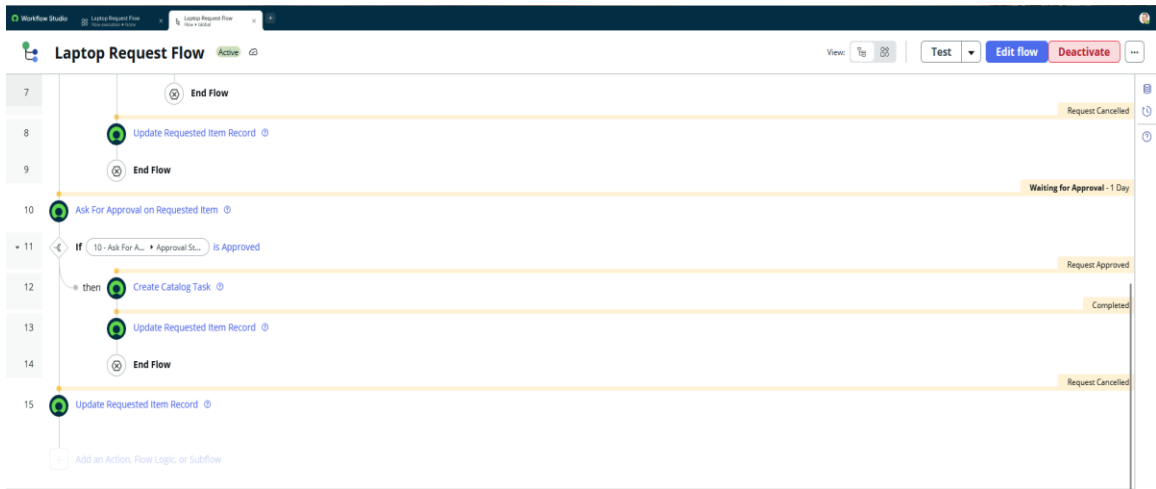
Step 4: Set request stages.

Step 5: Create catalog task.

Step 6: Assign task to IT support team.

Step 7: Complete fulfillment process.





Submitted Request - [RITM0010015](#)

servicenow | All | Favorites | History | Workspaces | Admin | Request New Laptop | Application scope: Global | Update url: Default | Contact

Service Catalog > Office > Services > Request New Laptop | Search catalog

Laptop Type
☐ Standard ☐ Gaming ☐ 2-in-1 ☐ Ultrabooks ☒ High Performance

Laptop Cost
☐ \$500 ☐ \$700 ☒ \$1000 ☐ \$1200

Business Justification

Order this item
Quantity: 1
Delivery time: 1 Day
[Order Now](#)
[Add to Cart](#)
Shopping Cart: Empty

servicenow | All | Favorites | History | Workspaces | Admin | Order Status: REQ0010015 | Application scope: Global | Update url: Default | Contact

Order Status | [Back to Catalog](#) | [Continue Shopping](#) | [Home](#)

Thank you, your request has been submitted

Order Placed: 2024-05-22 06:26:25
Request Number: [REQ0010015](#)
Estimated Delivery Date of Complete Order: 2024-05-23

Description	Delivery Date	Stage	Price (ex.)	Quantity	Total
	2024-05-23	▶ ○ ○ ○ ○		1	
				Total	

[Back to Catalog](#) | [Continue Shopping](#) | [Home](#)

Catalog Tasks			
Approvers (3)			
Group approvals (1)			
Approval for = RITM0010015			
State	Approver	Comments	Created
Requested	Alejandro Mascall		2026-05-22 06:26:28
Requested	Alene Rabeck		2026-05-22 06:26:28
Requested	Abel Tuter		2026-05-22 06:26:28

Workflow Studio			
Laptop Request Flow			
EXECUTION DETAILS			
Show Action Details			
TRIGGER			
Catalog Item Requested			
ACTIONS			
1	Get Catalog Variables from Request New Laptop	Core Action	Completed
2	If 1 - Get Catalog ... - laptop_s... Is High Performance	Flow Logic	Evaluated - True
3	Ask For Approval	Core Action	Completed
4	If 3 - Ask For A... - Approval St... Is Approved	Flow Logic	Evaluated - True
5	Create Catalog Task	Core Action	Completed
6	Update Record	Core Action	Completed
7	End	Flow Logic	Completed
8	Update Record	Core Action	Not Run
9	End	Flow Logic	Not Run
10	Ask For Approval	Core Action	Not Run

8. Dynamic Approval Logic

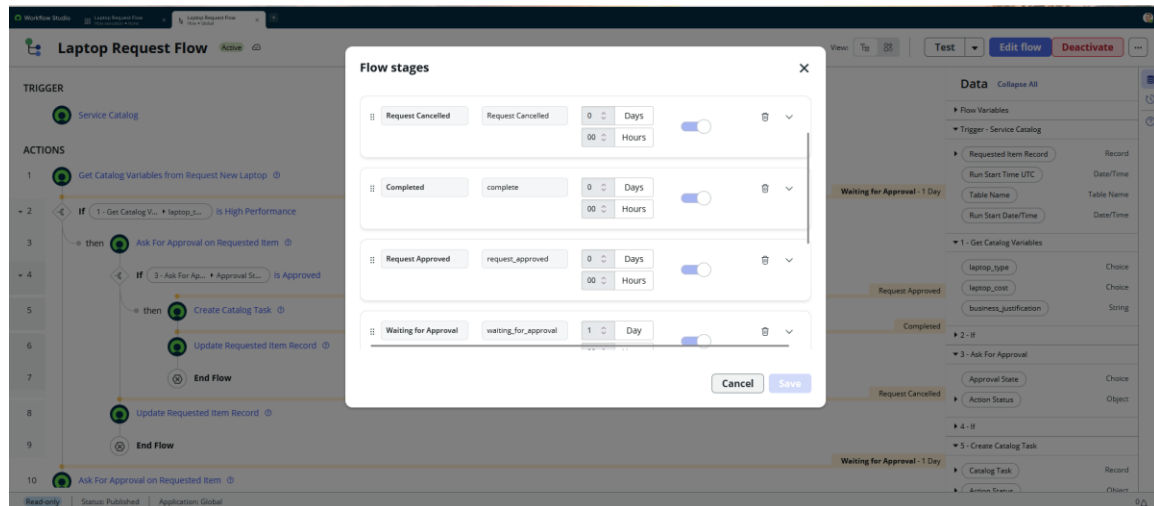
High-performance laptops require Manager and IT approval, while standard laptops require only Manager approval.

Catalog Tasks (1)			
Approvers (3)			
Group approvals (1)			
Approval for = RITM0010015			
State	Approver	Comments	Created
No Longer Required	Alejandro Mascall		2026-05-22 06:26:28
Approved	Alene Rabeck		2026-05-22 06:26:28
Approved	Abel Tuter		2026-05-22 06:26:28

9. Request Stages

Stages implemented:

- Request Submitted
- Approval in Progress
- Fulfillment
- Completed



10. Catalog Task Generation

After approval, catalog tasks are generated automatically and assigned to the IT Hardware Support Team.

5

Create Catalog Task

Core Action

Completed

2025-05-22 06:27:14

221ms

Configuration Details

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION	TYPE
Table Name	sc_task	sc_task	Table Name
Requested Item	RITM0010015	Trigger -> Requested Item ->	Reference
Short Description	Task Created for new High Performance Laptop	Task Created for new High Performance Laptop	String
Fields	assignment_group=5f6441efc0a8010e0177fcb589156352	assignment_group={"display":"IT Securities","value":"5f6441efc0a8010e0177fcb589156352","sys_id":"5f6...	Template Value
Wait	true	1	True/False
Template Catalog Item	Request New Laptop	Request New Laptop	Reference
Catalog Variables	laptop_type laptop_cost business_justification	laptop_type laptop_cost business_justification	Slush Bucket

Output Data

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION	TYPE
Action Status	{"Action Status":{"code":"0","message":"Success"}}		Object
Catalog Task	SCTASK0010013	task	Document ID
Don't Treat as Error	true	true	True/False

No Logs

11. Business Benefits

- Faster request processing
- Reduced manual work
- Improved request visibility

12. Conclusion

The IT Asset Request Automation project successfully demonstrates how ServiceNow can be used to automate and streamline enterprise IT service management processes. Using ServiceNow Flow Designer and Service Catalog, the project automates the complete lifecycle of IT asset requests, starting from employee request submission to approval management, catalog task generation, and final fulfilment by the IT support team.

The implementation of dynamic approval workflows, catalog variable retrieval, request stage tracking, and automated task assignment